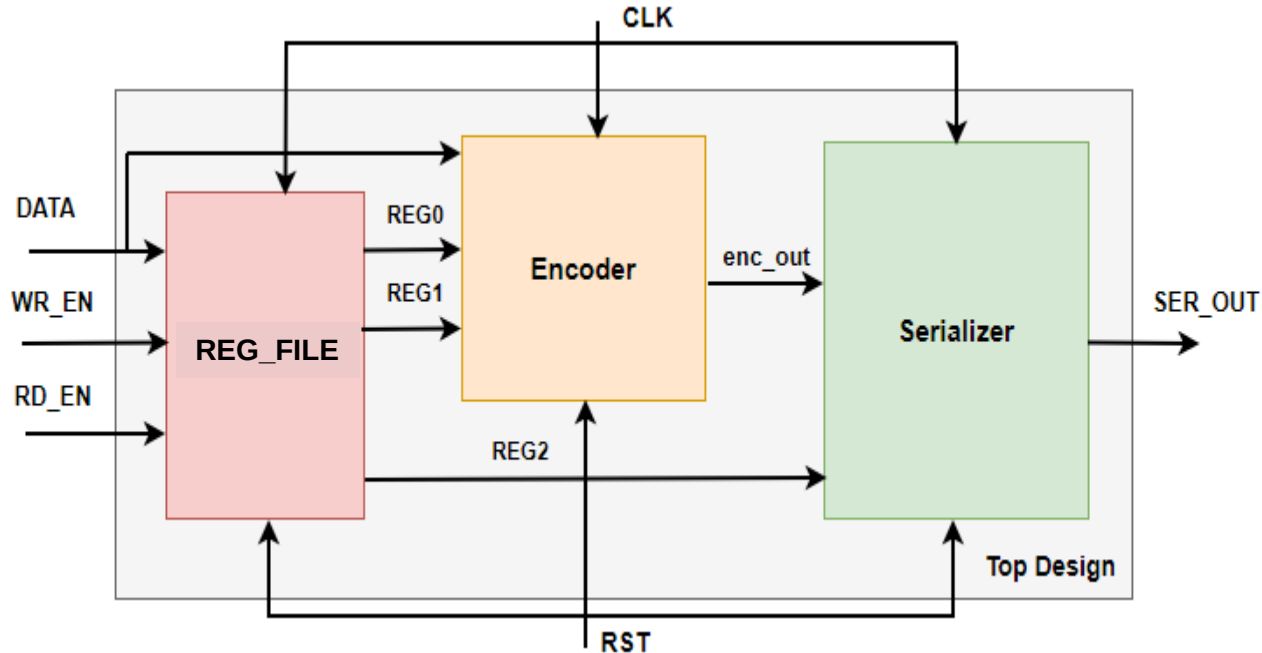


Write the Top Verilog Module

Structural Verilog is used to :-

1) Assemble several blocks together in a hierarchical design



Top_Design Ports:-

Inputs :-

- 1) DATA (8 bits)
- 2) WR_EN (1 bit)
- 3) RD_EN (1 bit)
- 4) CLK (1 bit)
- 5) RST (1 bit)

Outputs :-

- 1) SER_OUT (1 bit)

Top_Design Internal wires

- 1) REG0 (8 bits)
- 2) REG1 (8 bits)
- 3) REG2 (8 bits)
- 4) Enc_out (8 bit)

Write the Top Verilog Module

```
module Top_Design (  
  
    // Define input & output ports  
  
);  
  
    // Define internal connections  
  
    REG_FILE U_regfile (  
        .rst(),  
        .clk(),  
        .data(),    // connected to top module input port "DATA"  
        .wr_en(),   // connected to top module input port  
        .rd_en(),   // connected to top module input port  
        .reg0(),    // connected to encoder  
        .reg1(),    // connected to encoder  
        .reg2()     // connected to serializer  
    );
```

```
    Encoder U_encoder (  
        .rst(),  
        .clk(),  
        .data(),    // connected to top module input port "DATA"  
        .reg0(),    // connected to register file  
        .reg1(),    // connected to register file  
        .encoded_out() // connected to serializer  
    );  
  
    Serializer U_serializer (  
        .clk(),  
        .rst(),  
        .reg2(),    // connected to register file  
        .ser_in(),   // connected to encoder  
        .ser_out()   // connected to top module output port "SER_OUT"  
    );  
  
endmodule
```